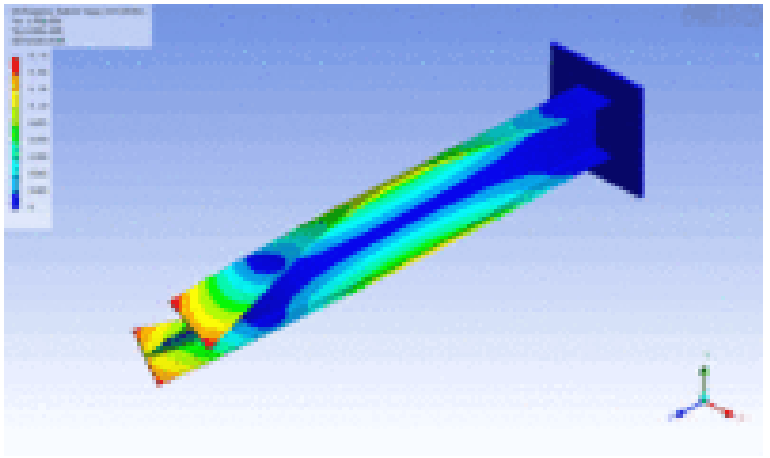
Chapter 5 Torsion

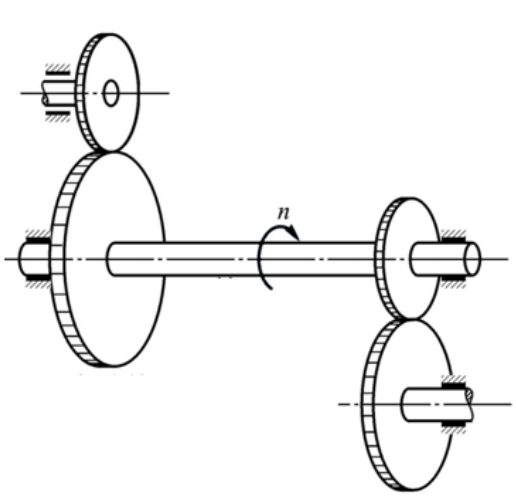


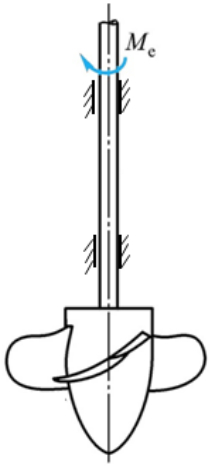
5.1 Introduction

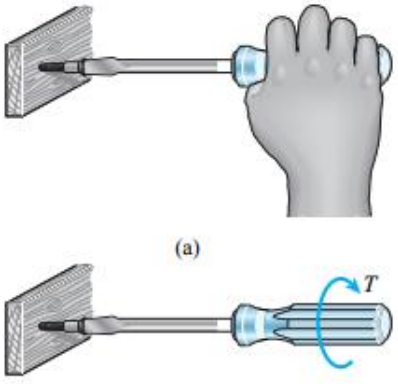
So far, we have looked primarily at normal stress and normal strain in columns and bars.

There are however many different types of loading.

For example, the torsion of a shaft. Shafts are encountered in many everyday situations. They are in most situations where a movement in one place is required to be transferred to another place.







5.2 The external torques in a shaft

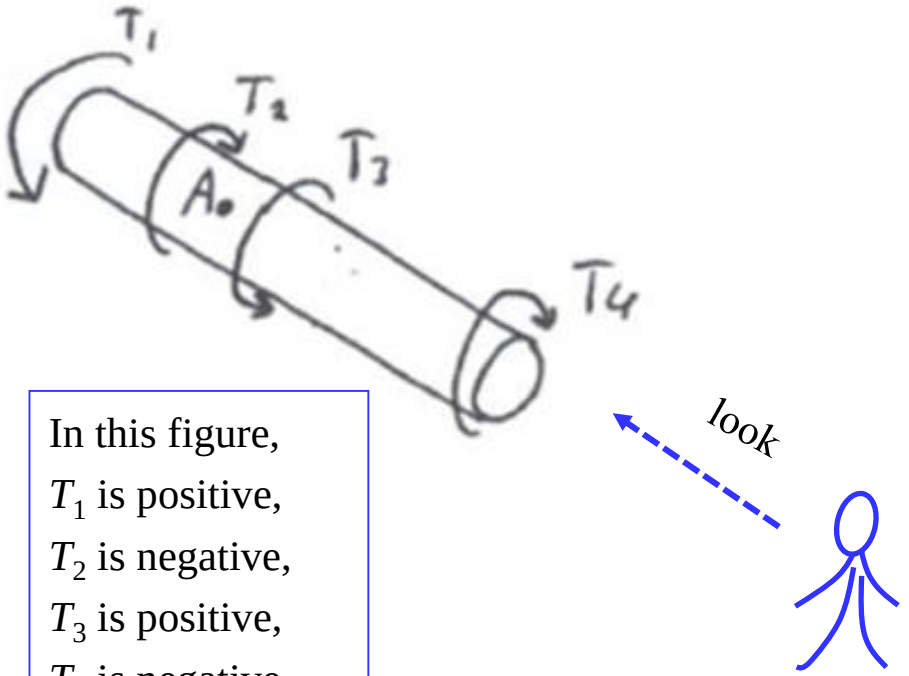
Torque is a moment that tends to twist a member about its longitudinal axis. Its effect is of primary concern in the design of driving shafts used in vehicles and machinery.

External torques are torques applied to a shaft.



Units of torque: Force.Length

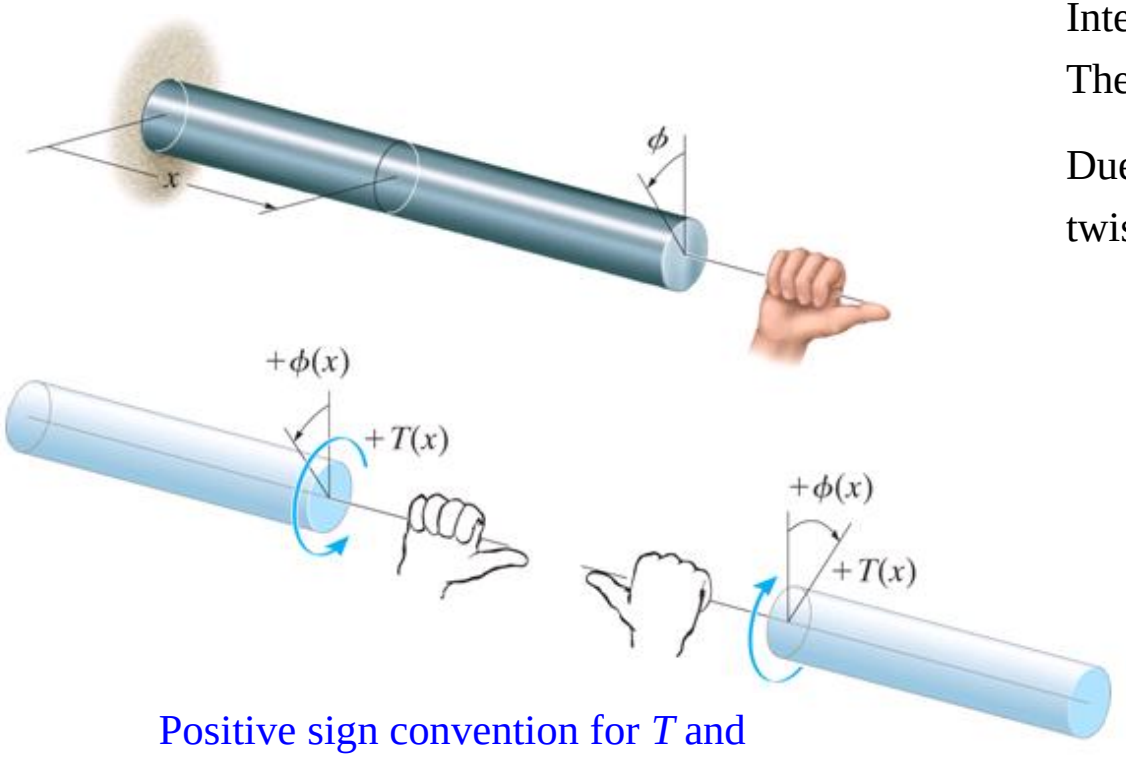
Here T_1, T_2, T_3 and T_4 are external torques.



The external torques are defined to be positive if, when you look at the shaft along its axial direction from an end, they are anti-clockwise.

In this figure,
T_1 is positive,
T_2 is negative,
T_3 is positive,
T_4 is negative.

5.3 Sign convention for internal torque (T) and angle of twist (ϕ)

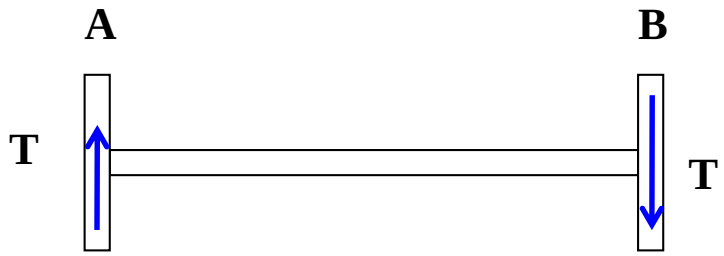


Positive sign convention for T and ϕ (angle of twist)

Internal torques are torques inside a shaft. They are torques carried by the shaft.

Due to an internal torque, the shaft will twist by an angle of twist, ϕ.

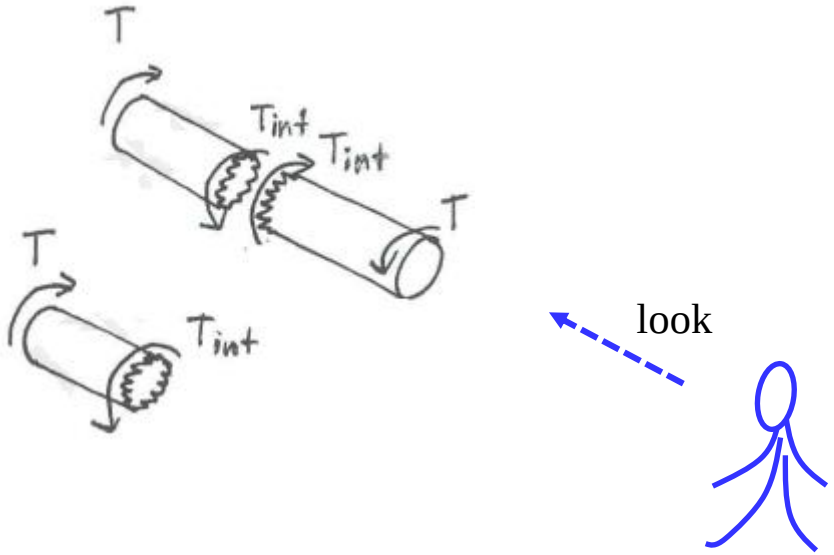
The direction of the torque is indicated by the *right-hand-curl* rule, namely, using your right hand, let your fingers curl in the direction of the torque, and then your thumb will point in the direction of the shaft (Namely, the thumb directs outward from the shaft).



When the shaft is viewed from the side, the **internal torque *T*** and ϕ are positive if, the torque at the left end appears to act upward, and the torque at the right end appears to act downward.

5.4 To determine the internal torque in a shaft

To determine the internal torque in the shaft, we cut the shaft somewhere between the applied torques. We then draw a **free body diagram** of the portion of the shaft on one side of the cut.



$$\sum M = 0$$

$$-T + T_{\text{int}} = 0$$

$$T_{\text{int}} = T$$

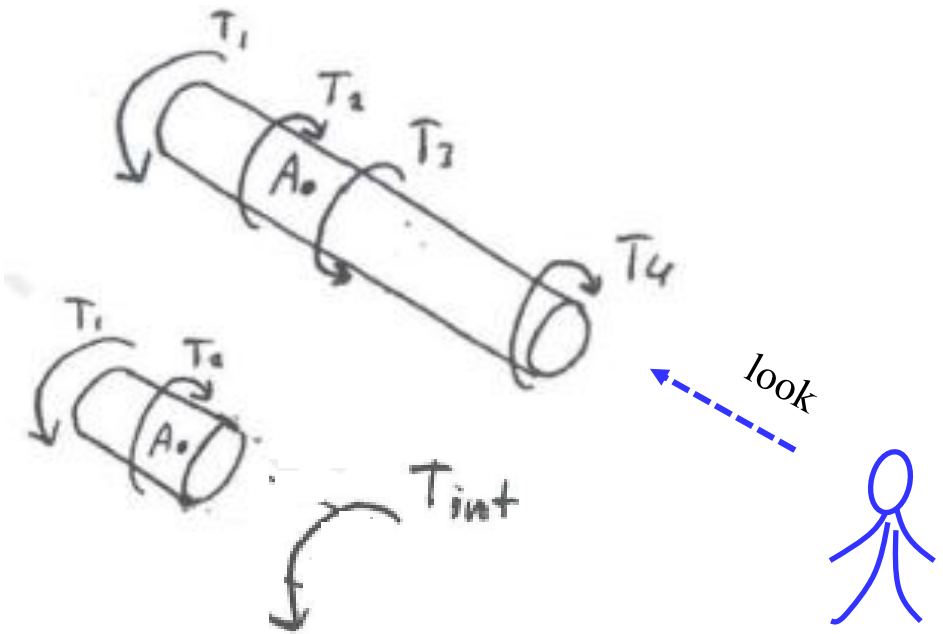
Here T is an external torque. T_{int} is the internal torque.

If we have multiple torques applied to the shaft, the internal torque being carried by the shaft will vary at different points along the shaft. In this situation, we have to take a free body diagram in each section of the shaft. For example, to find the internal torque at point A, we cut the shaft at A and draw a free body diagram of the shaft to the left of A.

$$\sum M = 0$$

$$T_1 - T_2 + T_{\text{int}} = 0$$

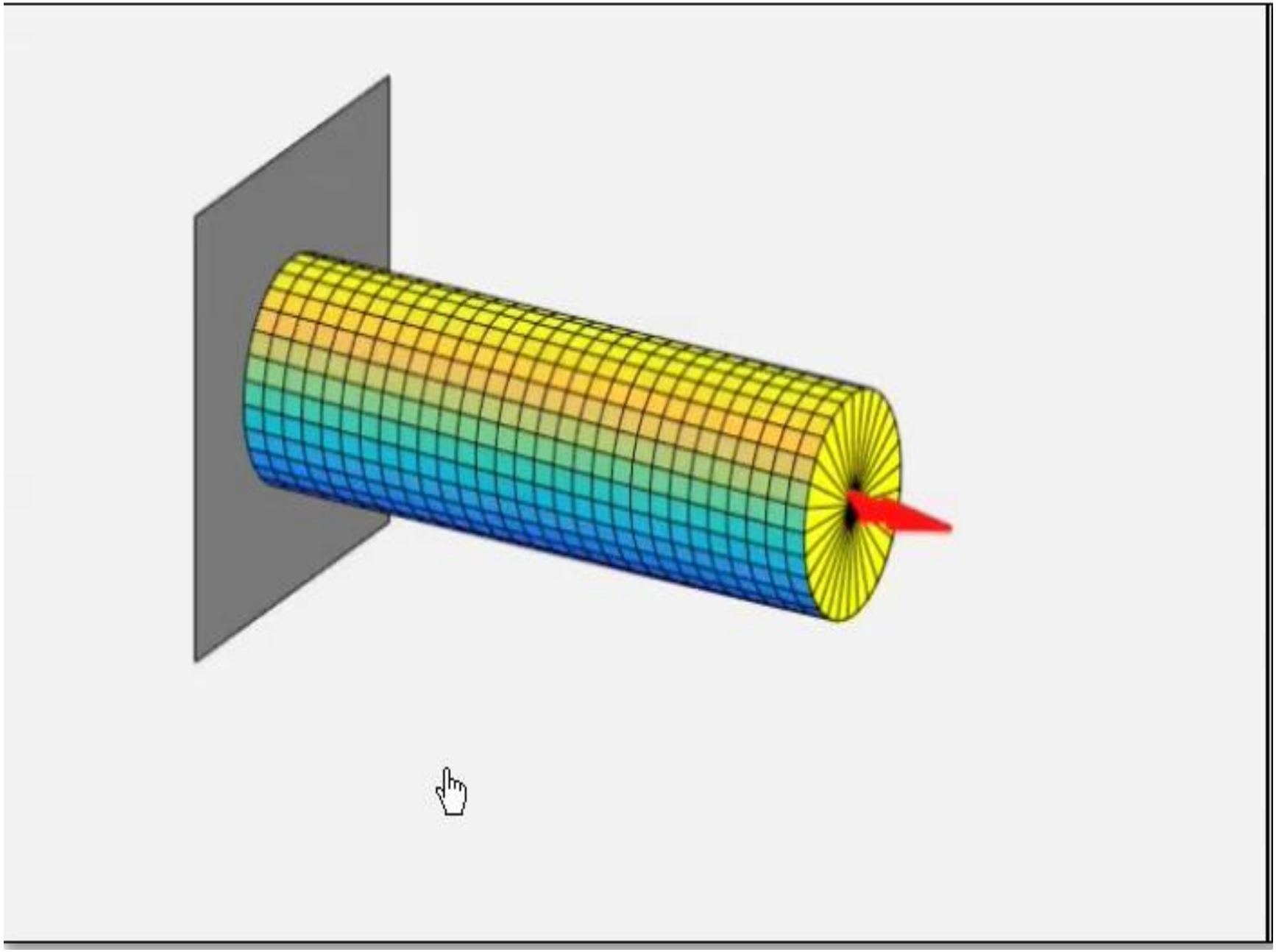
$$T_{\text{int}} = T_2 - T_1$$



Here T_1, T_2, T_3 and T_4 are external torques.

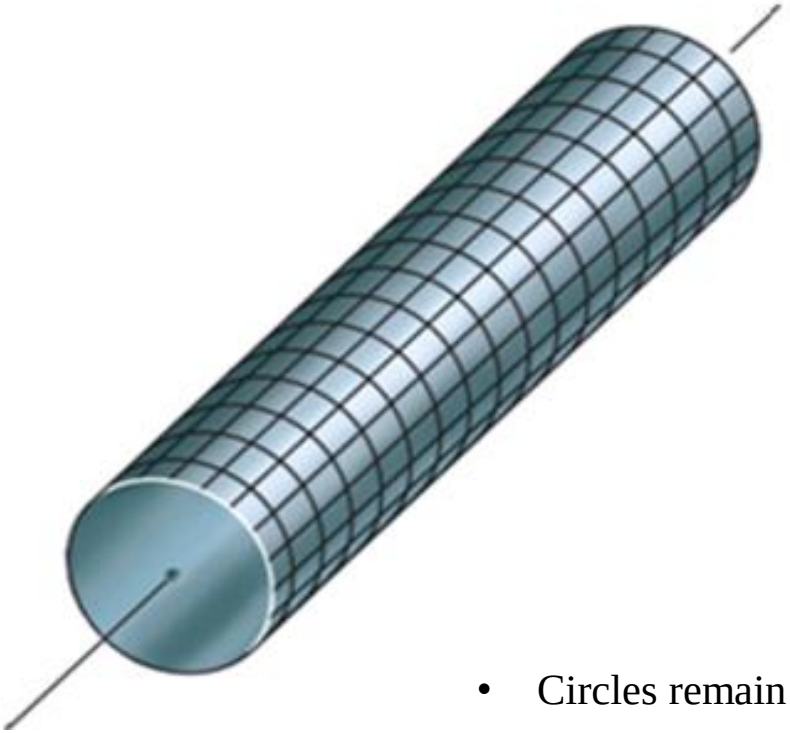
T_{int} is the internal torque.

5.4 Deformation assumption

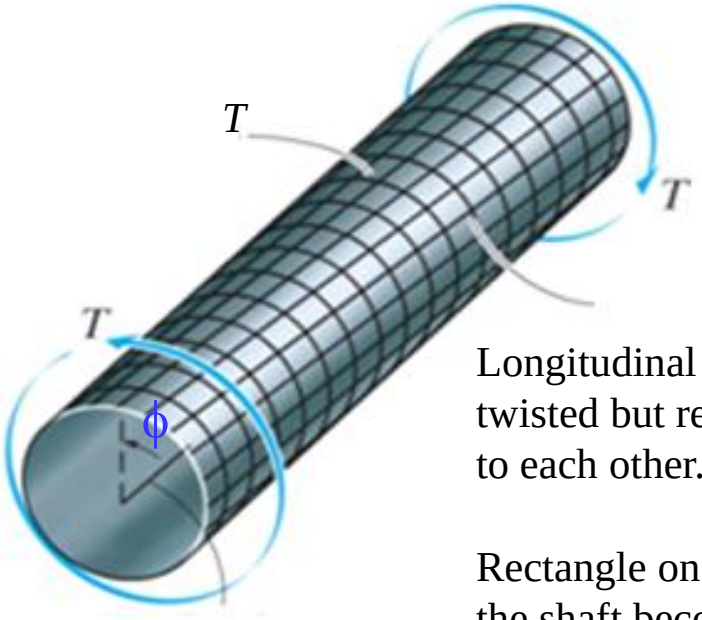


What we have observed ?

Original configuration



Deformed configuration



Longitudinal lines become twisted but remain parallel to each other.

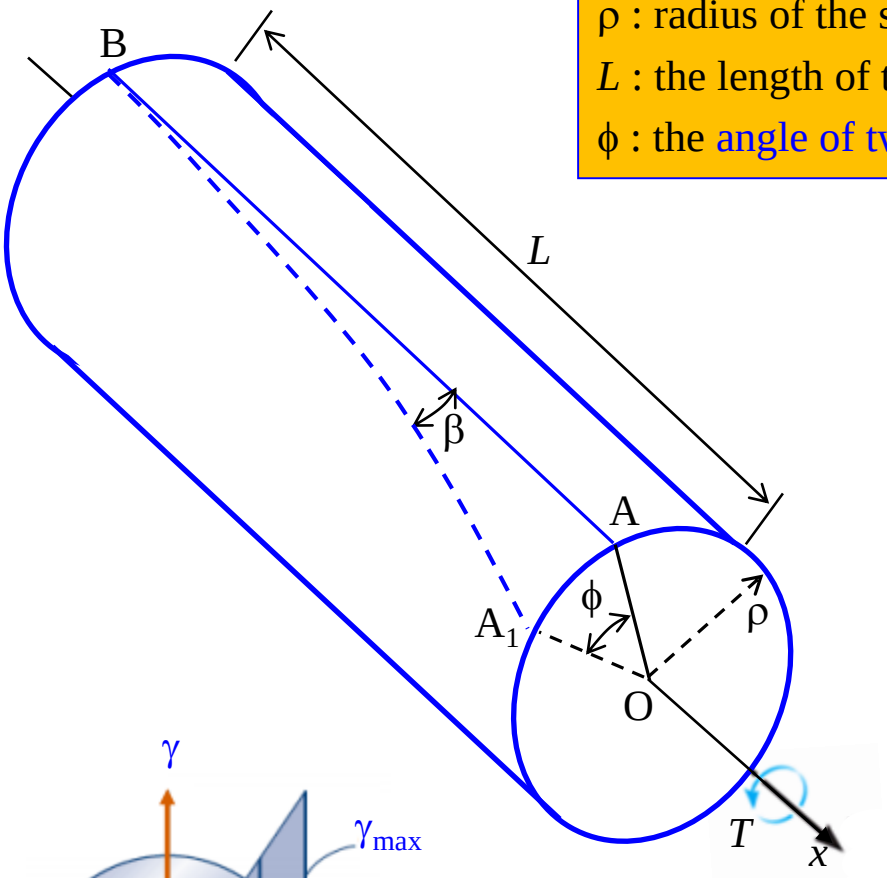
Rectangle on the surface of the shaft become a parallelogram

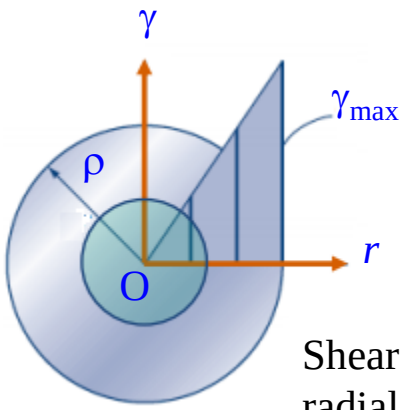
- Circles remain circular
- Radial lines remain straight and rotates.
- The cross section remains plane but rotates an angle of twist ϕ.

Assumptions: Linear, small and elastic deformation

5.5 Relating shear strain in a shaft to its angle of twist (circular bar)

ρ : radius of the shaft;
L : the length of the shaft
ϕ : the angle of twist under the internal torque *T*.





Shear strain varies linearly along each radial line of the cross-section

* Geometry relationship

$$\overline{AA_1} = \rho\phi$$

$$\overline{AA_1} = L\beta$$

$$\Rightarrow \quad \rho\phi = L\beta \qquad (5.1)$$

* Shear strain on the surface of the shaft:

$$\gamma_{\text{on the surface}} = \beta = \frac{\rho\phi}{L} \qquad (5.2)$$

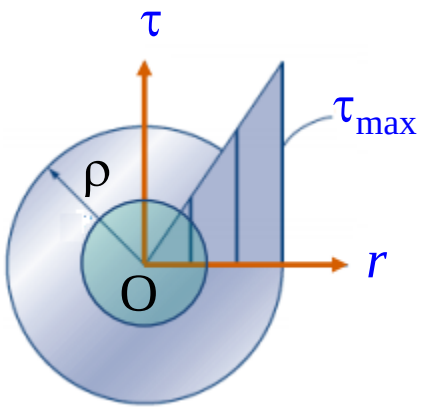
* Shear strain at the radius of *r*:

$$\gamma_{\text{at radius } r} = \frac{r\phi}{L} \qquad (5.3)$$

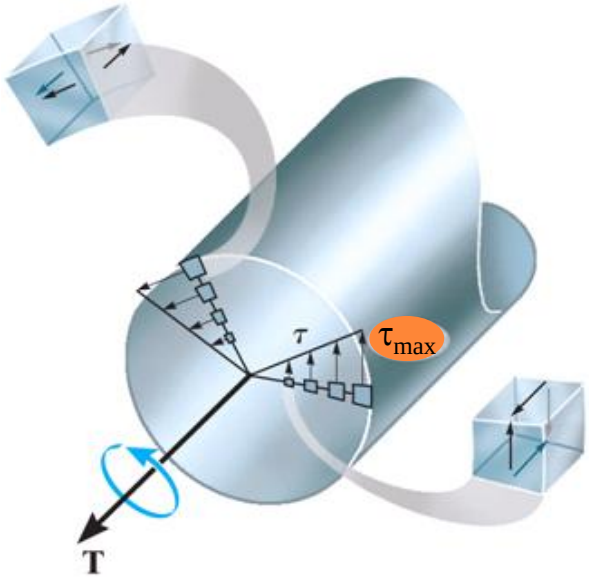
r is between zero and ρ

$$\gamma_{\text{at radius } r} = \frac{r\phi}{L} \xrightarrow{\tau = G\gamma} \tau = \frac{Gr\phi}{L} \qquad (5.4)$$

Hooke's Law, where G is the shear modulus, it is a material property.

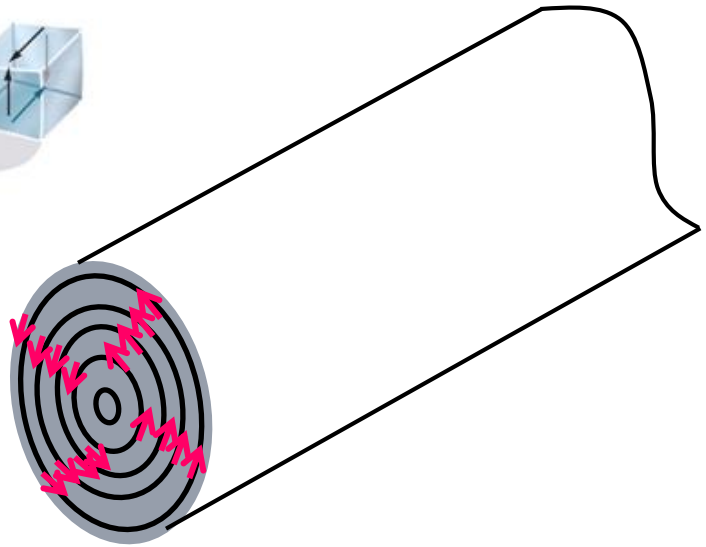


Linear distribution along r direction

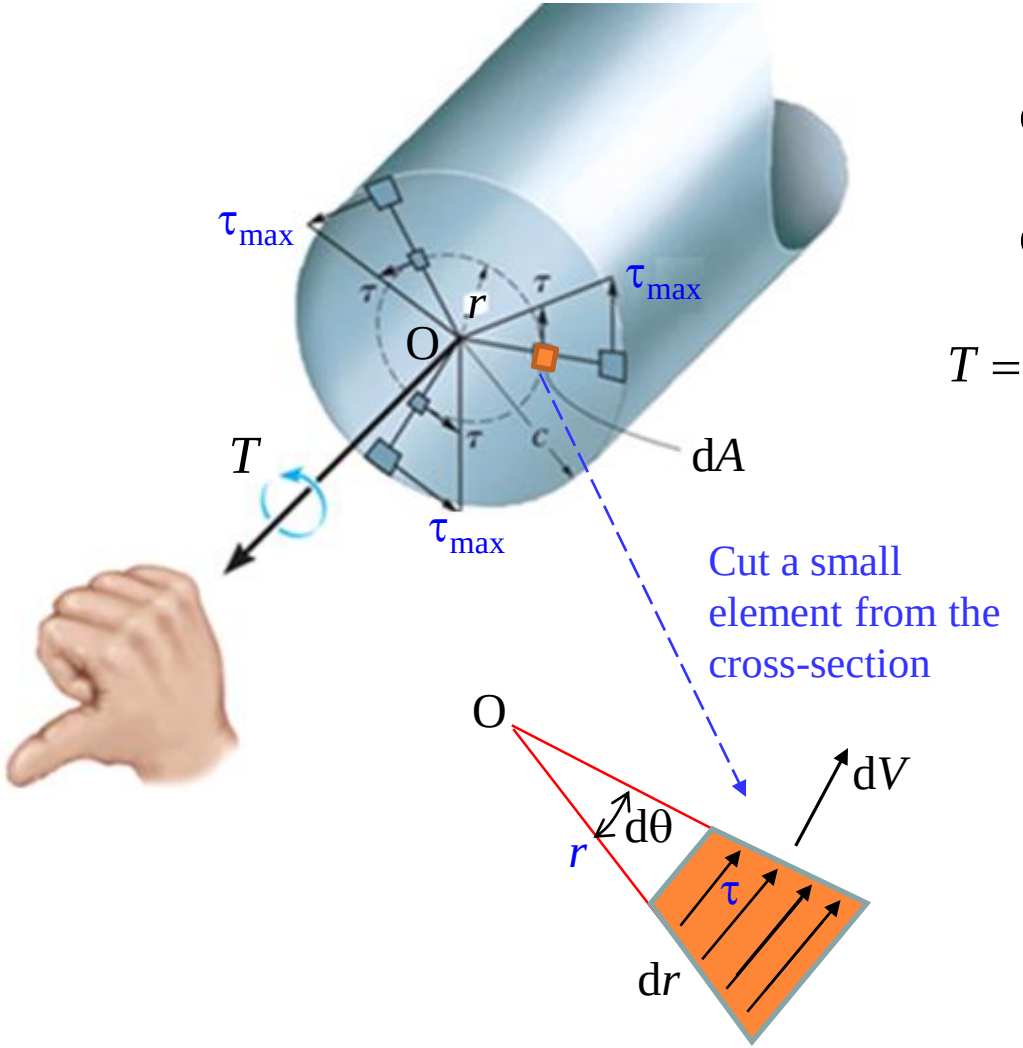


(1) Shear stress is parallel to circumferential direction;

(3) There is no shear stress in radial direction.



5.6. Toque – angle of twist relationship



$$dA = r d\theta dr$$

$$dV = \tau dA = \tau r d\theta dr$$

$$dT = dV \times r = \tau r^2 d\theta dr$$

$$T = \int_A dT = \int_A \tau r^2 dr d\theta = \int_A \frac{G r \phi}{L} r^2 dr d\theta = \frac{G \phi J}{L}$$

Thus

$$\phi = \frac{TL}{GJ} \qquad (5.5)$$

$$J = \int_A r^3 dr d\theta \qquad (5.6)$$

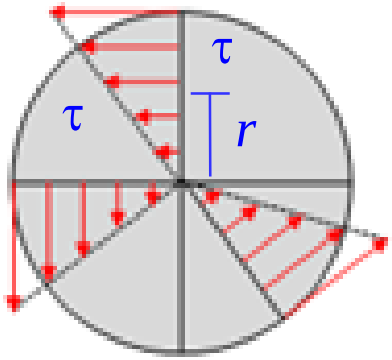
J is the polar moment of inertia of the shaft. It is a geometric property of the cross-section of the shaft.

- dV : shear force on the small element;
- dA : area of the small element;
- O: center of the cross-section;
- r : distance between the small element and centre.

5.7 Relating the shear stress to the internal torque

From Eqs. (5.4) and (5.5), we get the shear stress at the radius *r* on the cross-section

$$\tau = \frac{Gr\phi}{L} = \frac{Gr}{L}\frac{TL}{GJ} = \frac{Tr}{J} \qquad (5.7)$$



where *r* is the radius at which we want to calculate the shear stress

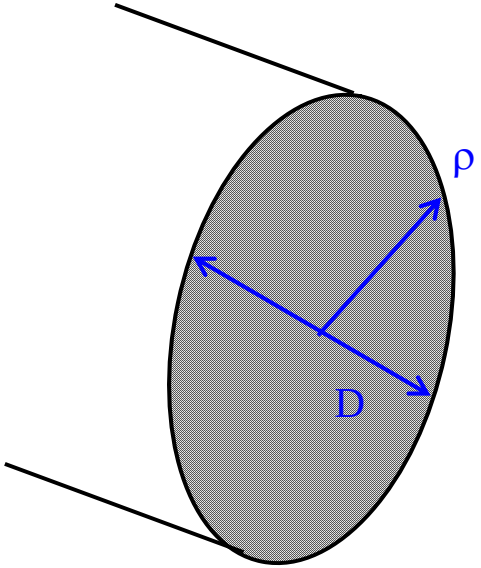
Notes:

1. In our analysis of the twisting of a shaft, the shaft must be circular (solid or hollow).

2. *T* is internal torque, and τ is the shear stress – don't confuse them.

3. We assume a small angle of twist (small deformation).

5.8 Polar moment of inertia of the shaft : *J*

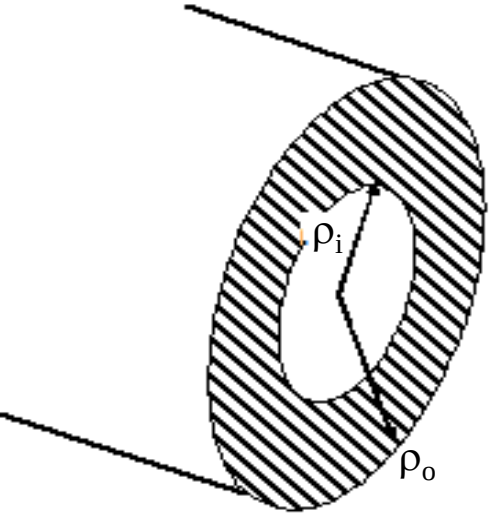


Solid shaft:

$$J = \int_A r^3 dr d\theta = \int_0^{\rho}\int_0^{2\pi} r^3 dr d\theta = \frac{\pi\rho^4}{2} = \frac{\pi D^4}{32}$$

ρ is the radius of the shaft. *D* is the diameter of the shaft.

Hollow shaft (e.g., a tube):



$$J = \int_A r^3 dr d\theta = \int_{\rho_i}^{\rho_o}\int_0^{2\pi} r^3 dr d\theta = \frac{\pi(\rho_o^4 - \rho_i^4)}{2}$$

ρ_o is the outer radius of the shaft.

ρ_i is the inner radius of the shaft.

5.9 Engineers' Theory of Torsion

Shear stress $\tau = \frac{Tr}{J} = \frac{Gr\phi}{L}$

Shear strain $\gamma = \frac{Tr}{GJ}$

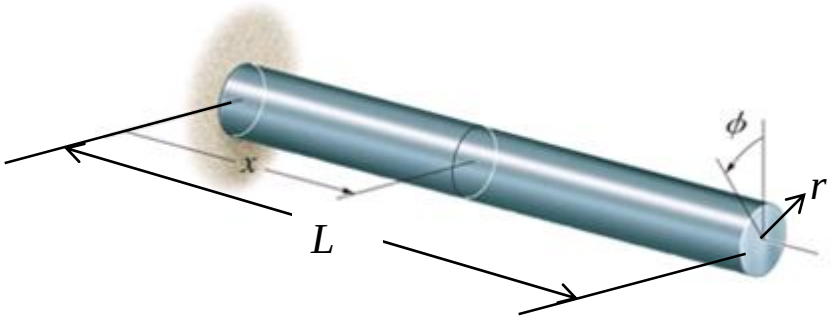
Angle of twist $\phi = \frac{TL}{GJ}$

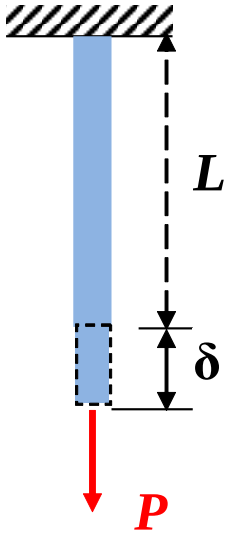
It is analogous to the axial loaded rod – the right

Normal stress $\sigma = \frac{P}{A} = \frac{E\delta}{L}$

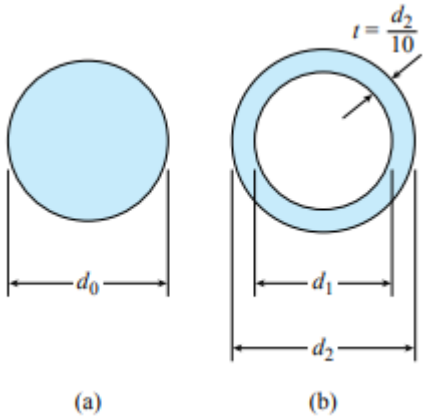
Normal strain $\varepsilon = \frac{P}{EA}$

Elongation $\delta = \frac{PL}{EA}$





Example 1. A steel shaft is to be manufactured either as a solid circular bar or as a circular tube (right figure). The shaft is required to transmit a torque of 1200 N.m without exceeding an allowable shear stress of 40 MPa.
(a)Determine the required diameter d_0 of the solid shaft.
(b)Determine the required outer diameter d_2 of the hollow shaft if the thickness t of the shaft is specified as one-tenth of the outer diameter.
(c)Determine the ratio of weights of the hollow and solid shafts.



Solution

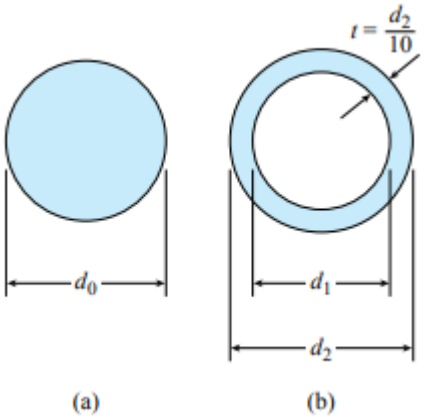
(1) Solid shaft

Maximum shear stress $\tau_{max} = \dfrac{T(d_0/2)}{J} = \dfrac{T(d_0/2)}{\dfrac{\pi d_0^4}{32}} = \dfrac{5.093T}{d_0^3}$

$$\tau_{max} < [\tau]$$

$$\Rightarrow d_0 \geq \sqrt[3]{\frac{5.093T}{[\tau]}} = \sqrt[3]{\frac{5.093(1200\times10^3)}{40}} = 53.5\ \text{mm}$$

(2) Hollow shaft

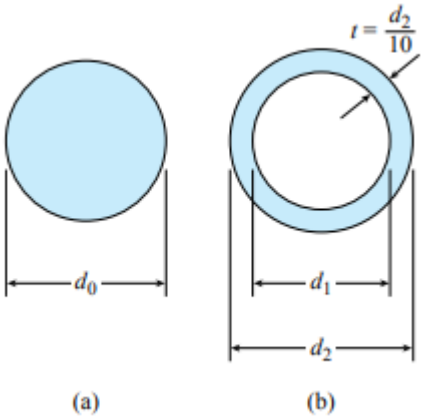


Maximum shear stress

$$\tau_{max} = \frac{T(d_2/2)}{J} = \frac{T(d_2/2)}{\frac{\pi(d_2^4 - d_1^4)}{32}} = \frac{5.093Td_2}{d_2^4 - (d_2 - 2t)^4} = \frac{5.093Td_2}{d_2^4 - (d_2 - 2\frac{d_2}{10})^4} = \frac{8.626T}{d_2^3}$$

$$\tau_{max} < [\tau]$$

$$\Rightarrow d_2 \geq \sqrt[3]{\frac{8.626T}{[\tau]}} = \sqrt[3]{\frac{8.626(1200 \times 10^3)}{40}} = 63.7 \text{ mm}$$



(3) Since the weights of the shafts are proportional to their cross-sectional areas, we can express the ratio of the weight of the hollow shaft to the weight of the solid shaft as follows:

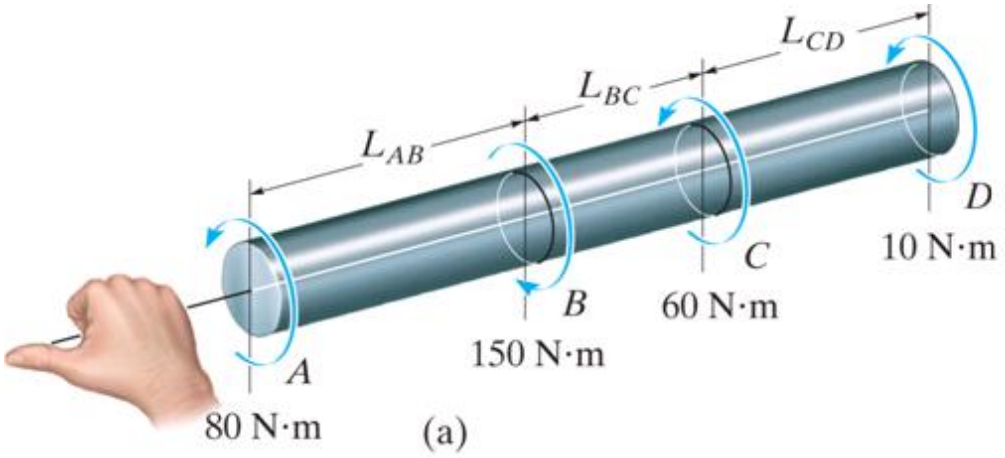
$$d_1 = d_2 - 2t = d_2 - 2\frac{d_2}{10} = 63.7 - \frac{63.7}{5} = 51.0\ \text{mm}$$

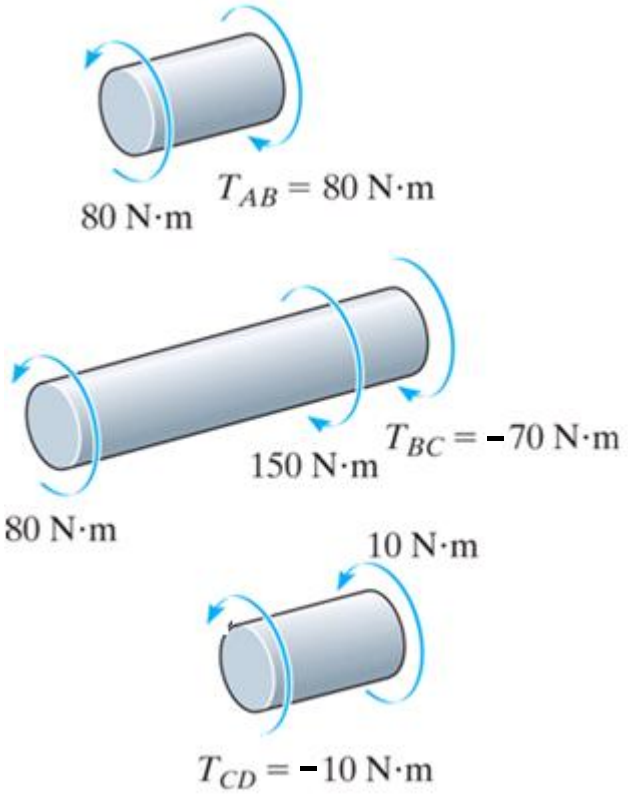
$$\frac{W_{\text{hollow}}}{W_{\text{solid}}} = \frac{A_{\text{hollow}}}{A_{\text{solid}}} = \frac{\frac{\pi}{4}(d_2^2 - d_1^2)}{\frac{\pi}{4}d_0^2} = \frac{63.7^2 - 51.0^2}{53.5^2} = 0.51$$

These results show that the hollow shaft uses only 51 % as much material as does the solid shaft.

Note: This example illustrates how to determine the required sizes of both solid bars and circular tubes when allowable stresses are known. It also illustrates the fact that circular tubes are more efficient in the use of materials than are solid circular bars.

5.10 Multiple torques





In order to determine the magnitude and sign of the internal torque at a point, we cut the shaft at the point and draw a free body diagram of the portion on one side of the cut.

We draw the unknown internal torque in the positive direction (anticlockwise). The negative values we obtain indicate that they act clockwise.

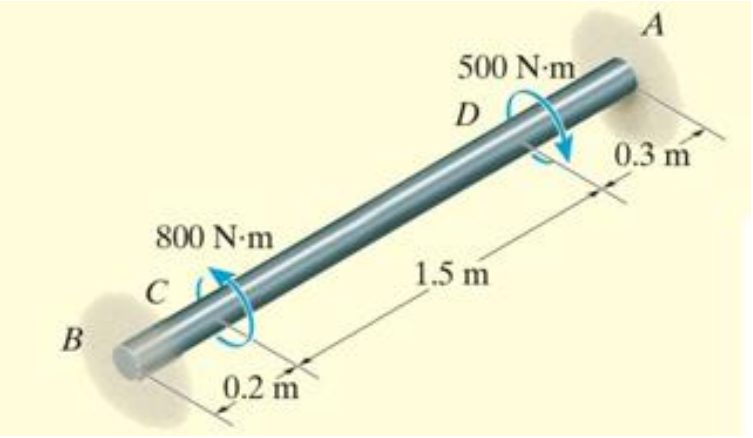
We can use the torques to calculate the angle of twist of a shaft:

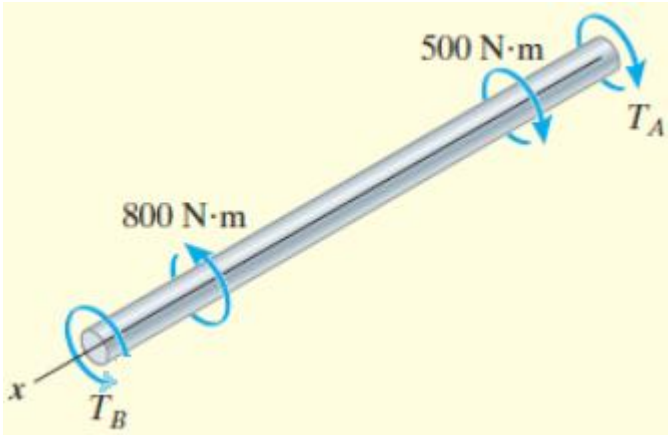
$$\phi_{A/D} = \phi_{A/B} + \phi_{B/C} + \phi_{C/D} = \frac{T_{A/B}L_{AB}}{G_{AB}J_{AB}} + \frac{T_{B/C}L_{BC}}{G_{BC}J_{BC}} + \frac{T_{C/D}L_{CD}}{G_{CD}J_{CD}} = \sum\frac{TL}{GJ} = \sum\frac{TL}{GJ} = \sum\frac{TL}{GJ}$$

5.11 Indeterminate torque – loading members

When studying the indeterminate problems in torque, in addition to the **equilibrium equation(s)**, the **compatibility condition** and **material behavior** must be used.

Example 2. The solid steel shaft shown in the figure has a diameter of 20 mm. It is subjected to two torques, determine the reactions at the fixed supports A and B.





Solution : Divide the shaft into three segments, BC, CD, and DA, based on the loading conditions. Reactions at the fixed ends A and B are denoted as T_A and T_B, respectively.

(1) Equilibrium equation:

$$T_B + 800 - 500 - T_A = 0$$

(2) Compatibility condition:

$$\phi_{B/A} = \phi_{B/C} + \phi_{C/D} + \phi_{D/A} = 0$$

(3) Angle of twist – torque relationships:

$$\phi_{B/C} = \frac{T_B \overline{BC}}{GJ}, \qquad \phi_{C/D} = \frac{(T_B + 800)\overline{CD}}{GJ},$$

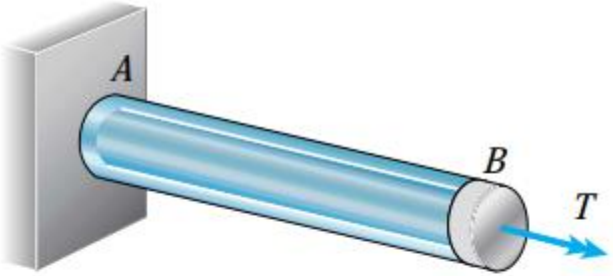
$$\phi_{D/A} = \frac{(T_B + 800 - 500)\overline{DA}}{GJ}$$

From (2) and (3), one gets

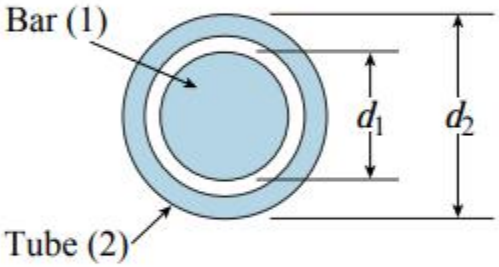
$$\frac{T_B \times 0.2 + (T_B + 800) \times 1.5 + (T_B + 800 - 500) \times 0.3}{GJ} = 0$$

Solving, $T_B = -645$ N.m

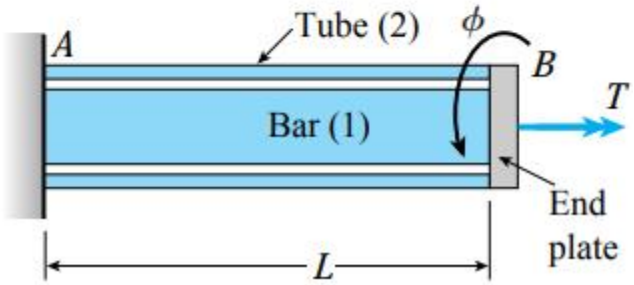
From (1), $T_A = -345$ N.m



(a)



(b)



(c)

Example 3: The composite bar AB shown in Fig. a is attached to a fixed support at end A and loaded by a torque T at end B. The bar consists of two parts: a solid bar and a tube (Figs. b and c), with both the solid bar and the tube joined to a rigid end plate at B. Determine: (1) the angle of twist of the composite bar, and (2) the maximum shear stress in the bar and the tube.

Solution: For convenience, we will identify the solid bar and tube (and their properties) by the numerals 1 and 2, respectively. For instance, the diameter of the solid bar is denoted d_1 and the outer diameter of the tube is denoted d_2. A small gap exists between the bar and the tube, and therefore the inner diameter of the tube is slightly larger than the diameter d_1 of the bar.

(1). Write the equilibrium equation:

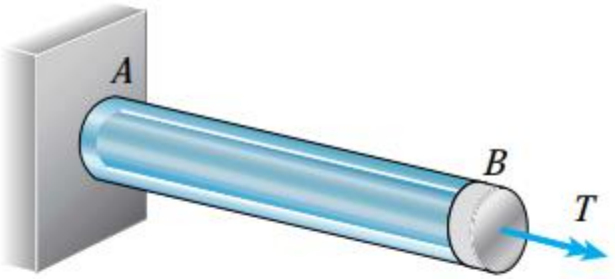
$$T_1 + T_2 = T$$

(2). Describe the compatibility condition:

$$\phi_1 = \phi_2 = \phi$$

(3). Give the material behavior (angle of twist – torque relationship):

$$\phi_1 = \frac{T_1 L}{(GJ)_{(1)}}, \quad \phi_2 = \frac{T_2 L}{(GJ)_{(2)}}$$

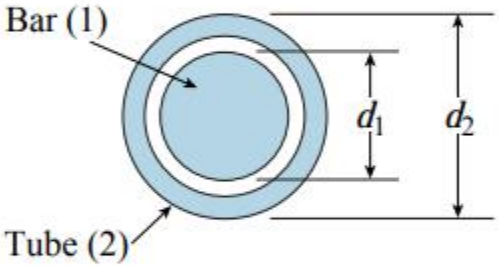


(a)

(4). Find the angle of twist. From (1) to (3), we get

$$\phi = \frac{TL}{(GJ)_{(1)} + (GJ)_{(2)}}$$

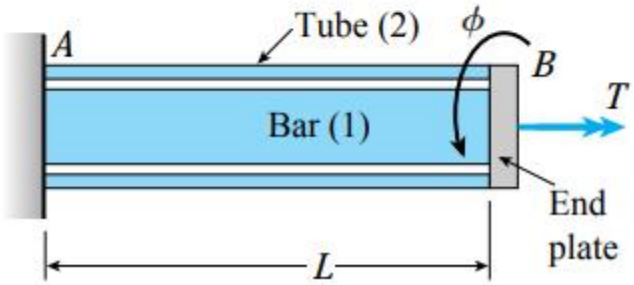
(5). Find the maximum shear stresses in the bar and tube, using Eq. (5.7)



(b)

In the bar, maximum shear stress is at $r = d_1/2$:

$$\tau_{\text{max in bar}} = \frac{Gr\phi}{L} = \frac{G_1 \dfrac{d_1}{2} \dfrac{TL}{(GJ)_{(1)} + (GJ)_{(2)}}}{L} = \frac{d_1}{2} \frac{TG_1}{(GJ)_{(1)} + (GJ)_{(2)}}$$



(c)

In the tube, maximum shear stress is at $r = d_2/2$:

$$\tau_{\text{max in tube}} = \frac{Gr\phi}{L} = \frac{G_2 \dfrac{d_2}{2} \dfrac{TL}{(GJ)_{(1)} + (GJ)_{(2)}}}{L} = \frac{d_2}{2} \frac{TG_2}{(GJ)_{(1)} + (GJ)_{(2)}}$$

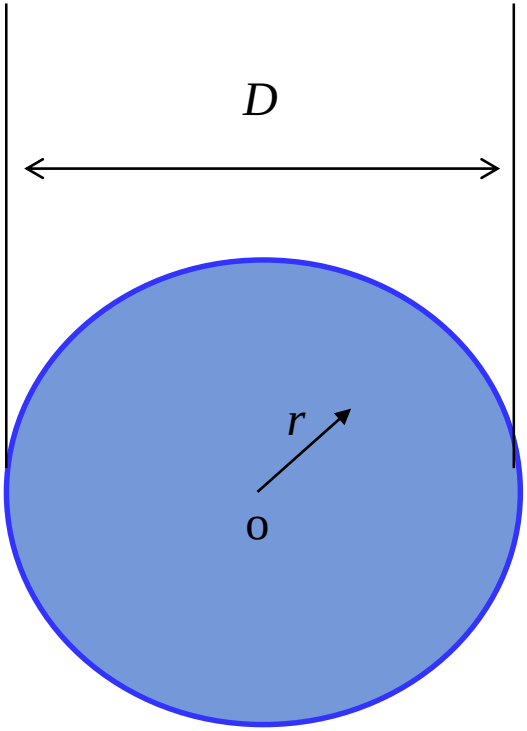
Torsion & Torque

Angle of twist

$$\phi = \frac{TL}{GJ}$$

Shear stress

$$\tau = \frac{Tr}{J} = \frac{Gr\phi}{L}$$



$$J = \frac{\pi D^4}{32}$$

- L is the length of the shaft.
- G is the shear modulus.
- J is the polar moment of area.
- r is the radius at which we want to calculate the shear stress.